

AI JOB REPLACEMENT ANALYSIS

2020-2026

Exploratory Data Analysis

OBJECTIVE

The primary objective of this project is to carry out a thorough Exploratory Data Analysis on the AI Job Replacement dataset in order to gain insights into how Artificial Intelligence is transforming the global workforce from 2020 to 2026.

The project aims to answer the following key research questions:

- What are the most automation-prone job roles?
- What industries and countries are most exposed to automation risk?
- What has been the year-over-year change in AI replacement scores, automation risk, and reskilling urgency?
- What is the relationship between skill gap index, salary change, and automation risk?
- Does skill gap for workers affect their level of automation risk?
- How does wage volatility vary across different risk categories?

The expected outcome would be a complete set of visualizations, insights, and an interactive Streamlit dashboard that clearly shows the trends, patterns, and risk areas of AI-driven job displacement in the global workforce.

DATA SOURCE

Dataset Overview

ATTRIBUTE	DETAILS
Dataset Name	ai_job_replacement_2020_2026.csv
Source	Kaggle
Total Records	15,000
Year Range	2020-2026
Countries	9(USA, UK, India, Canada, Germany, Australia, Brazil, Japan, Singapore)
Industries	8(Technology, Finance, Healthcare, Education, Manufacturing, Retail, Transportation, Energy)
Null Values	0
Duplicate Values	0

Key Columns

COLUMN NAME	DESCRIPTION
automation_risk_percent	Percentage likelihood of a job being automated
ai_replacement_score	Score indicating AI replacement probability (0–100+)
skill_gap_index	Gap between current worker skills and required skills
Salary_before_usd	Salary before AI adoption (in USD)
salary_after_usd	Salary after AI adoption (in USD)
salary_change_percent	Percentage change in salary due to AI adoption
reskilling_urgency_score	Urgency score for worker reskilling needs
wage_volatility_index	Index measuring salary instability in the role
remote_feasibility_score	Score indicating how feasibly a job can be done remotely
automation_risk_category	Categorical label — Low, Medium, or High automation risk

METHODOLOGY

This project follows a structured Exploratory Data Analysis (EDA) approach. The methodology is divided into the following steps:

- 1. Data Loading & Inspection**
The data was loaded into pandas, and initial inspection was done to understand the data, data types, and statistics.
- 2. Data Cleaning**
The data was clean, with no missing values or duplicates. No columns were dropped during this process. The column 'automation_risk_category' was confirmed to have valid data, with categories being 'Low,' 'Medium,' and 'High.'
- 3. Univariate Analysis**
Individual numeric columns were analysed through histograms to understand their distribution.
- 4. Bivariate Analysis**
Key metrics were compared across industries, countries, and risk categories through bar charts and horizontal bar charts.
- 5. Multivariate Analysis**
Violins and scatter plots were used to understand multi-variable relationships, especially concerning skill gap, AI replacement score, and risk categories.
- 6. Trend Analysis**
A heatmap was used to analyse how reskilling urgency changed across industries and years (2020–2026).

7. Correlation Analysis

A correlation heatmap was generated to identify relationships between all numeric features.

TOOLS AND TECHNIQUES

1. **Python** - Core programming language for all analysis and dashboard
2. **Pandas** - read_csv, groupby, mean, sort_values, pivot_table, value_counts
3. **NumPy** - Numerical operations and array handling
4. **Matplotlib** - plt.hist, plt.bar, plt.barh, plt.boxplot, plt.scatter, plt.show
5. **Seaborn** - sns.violinplot, sns.heatmap

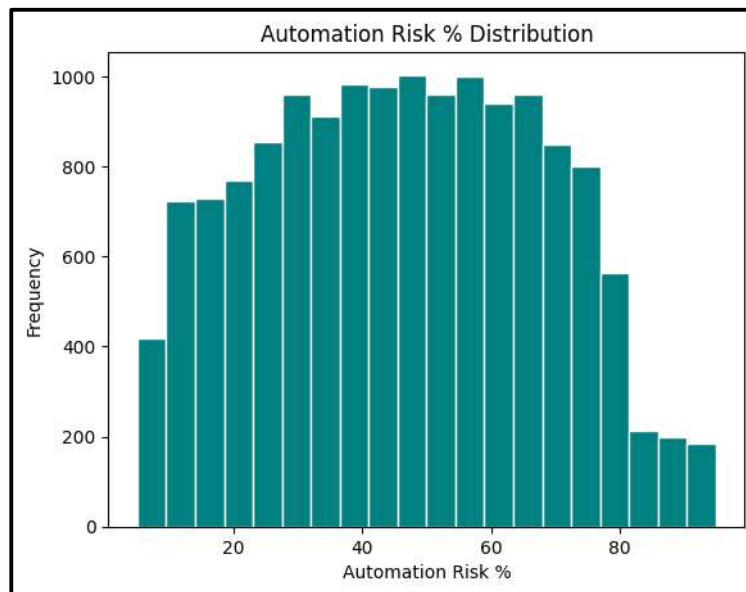
EDA SUMMARY STATISTICS

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EDA SUMMARY – AI Job Replacement	
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Total Records	: 15,000
Industries	: 8
Countries	: 9
Job Roles	: 10
Year Range	: 2020 → 2026
Mean Automation Risk	: 46.18%
Mean AI Replace Score	: 46.16
Mean Skill Gap Index	: 50.00
Mean Salary Change	: 0.11%
Mean Reskilling Urgency	: 35.87
High Risk Records	: 4,496 (30.0%)
Most At-Risk Role	: Truck Driver
Most At-Risk Industry	: Energy
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DATA VISUALIZATION

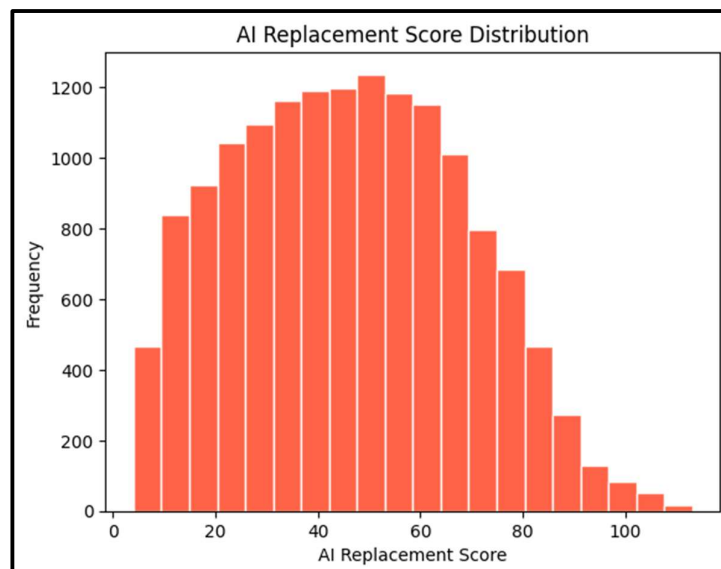
The following section provides detailed interpretation of each chart produced during the EDA, based on actual values computed from the dataset.

Automation Risk % Distribution



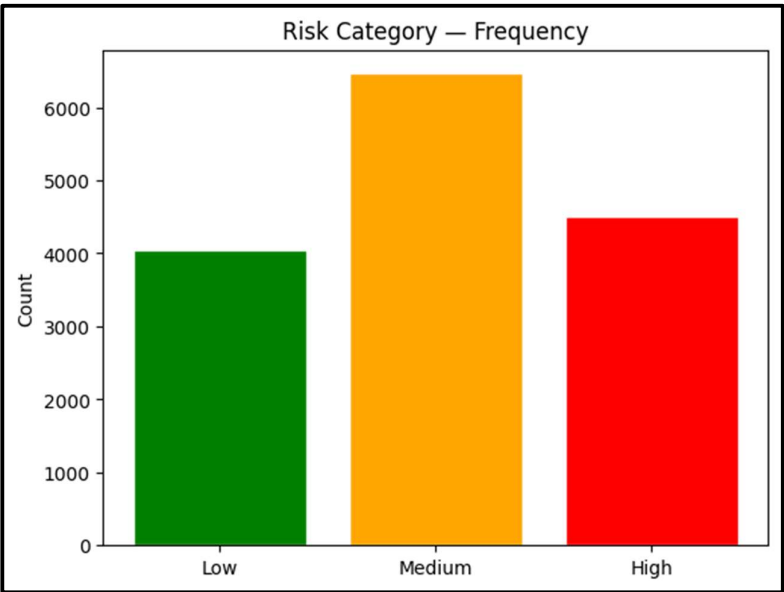
From this histogram, we observe that the distribution is slightly right-skewed because most jobs are concentrated in the range of 30-60% in terms of automation risk. The highest peak, which ranges from 40-50%, contains 1,981 records. Very few jobs are at extremely high risk, as indicated by only 384 jobs having more than 85% risk. The mean automation risk is 46.18%, and the median is 46.24%.

AI Replacement Score Distribution



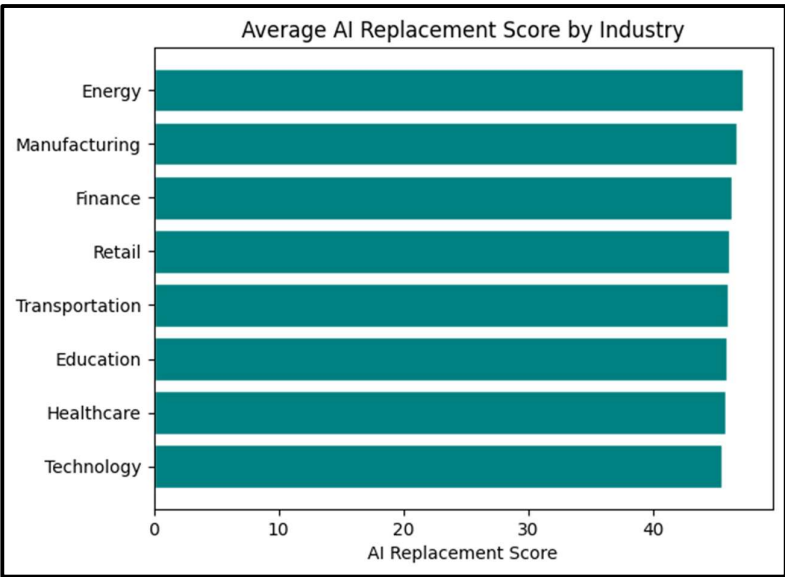
The AI replacement score also has a mean value of 46.16 and a median value of 45.68, closely resembling the distribution of automation risk. The range for this score is between 4.01 and 113.07, while 75% of jobs are below 62.71. The similarity between the mean and median values indicates symmetry in this distribution, except for jobs with extremely high risk.

Risk Category Frequency



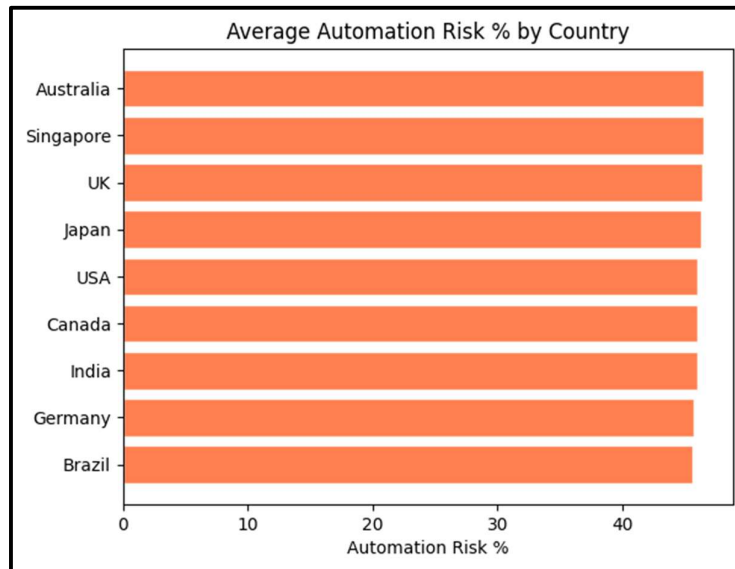
The data set is not equally distributed. The Medium risk group is dominant with 6,464 records (43.1%), followed by the High-risk group with 4,496 records (30.0%), and then the Low-risk group with 4,040 records (26.9%). The implication is that the majority of the jobs are in the moderate-to-high risk zone for automation, and this is a realistic reflection of the state of AI disruption in the world workforce today.

AI Replacement Score by Industry



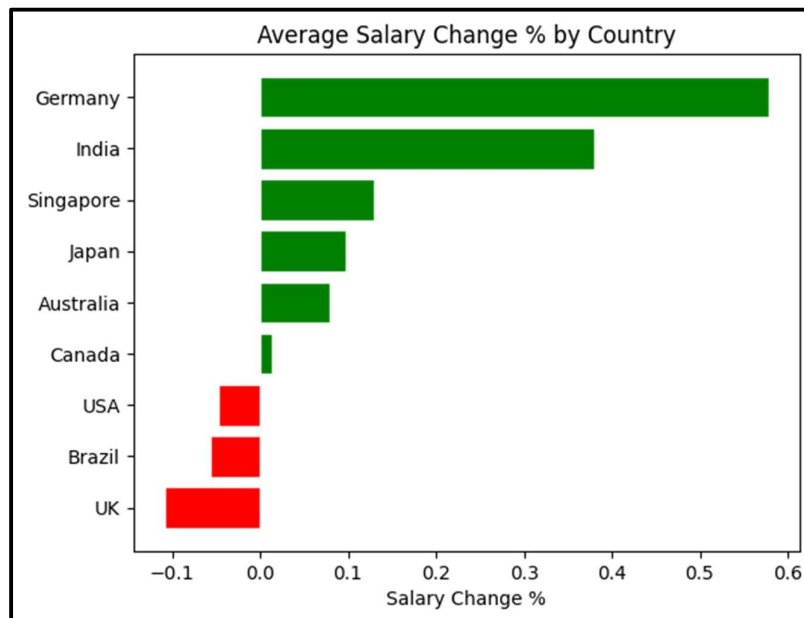
Unexpectedly, Energy (47.18) and Manufacturing (46.68) are in first place for average AI replacement, while Technology (45.50) is in last place, defying conventional wisdom and suggesting that these sectors are more exposed due to significant advancements in robotics technology driven by AI. The range for all industries is surprisingly small, from 45.5 to 47.2, suggesting that all industries are being impacted by AI to some extent.

Automation Risk % by Country



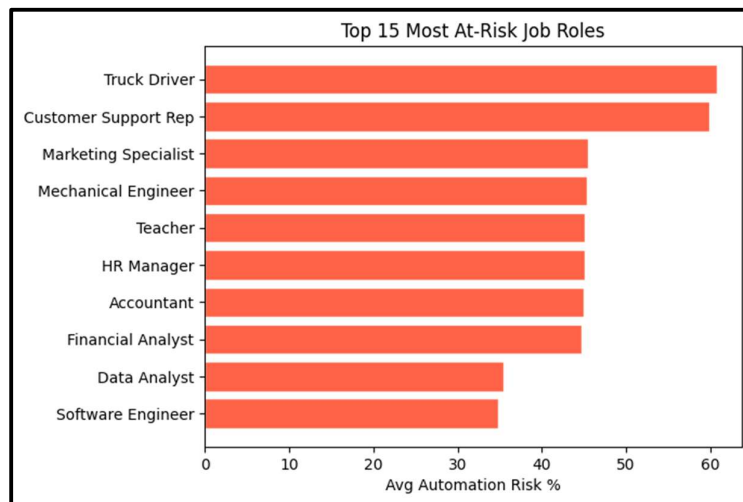
All nine countries display almost identical automation risk percentages ranging from 45.6% to 46.6%, with Australia having a slight edge at 46.57%, while Brazil, on the other hand, has the lowest at 45.61%. This slight 1% differential in all the countries proves that AI disruption is a universal trend worldwide, and none of these nations are at a disadvantage or advantage in this regard.

Salary Change % by Country



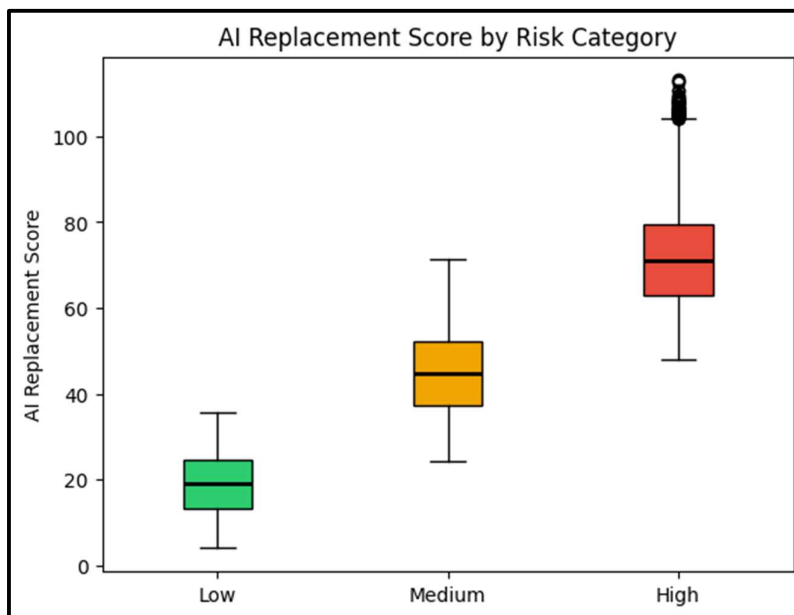
Contrary to expectations, most countries have a positive change in salary. Germany ranks highest in terms of salary change at +0.58%, followed by India at +0.38%, and Singapore at +0.13%. The only countries that have negative values are UK (-0.11%), Brazil (-0.06%), and USA (-0.05%). It appears that AI is improving productivity and salary in most countries.

Top 15 Most At-Risk Job Roles



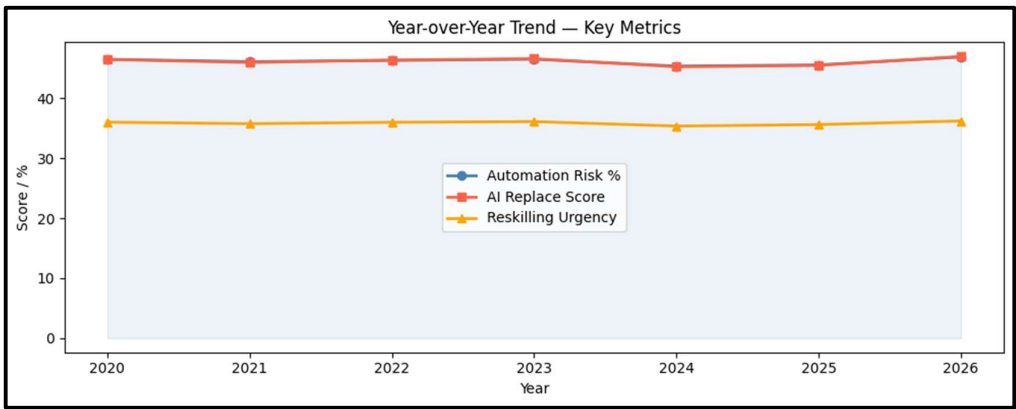
Truck Driver (60.74%) is the most at-risk occupation. The second is Customer Support Rep (59.85%). White-collar jobs such as Marketing Specialist (45.44%), Mechanical Engineer (45.40%), and Teacher (45.14%) make up the top 10. The least at-risk occupations are Software Engineer (34.85%) and Data Analyst (35.46%), which fall under programming and analysis.

AI Score by Risk Category — Box Plot



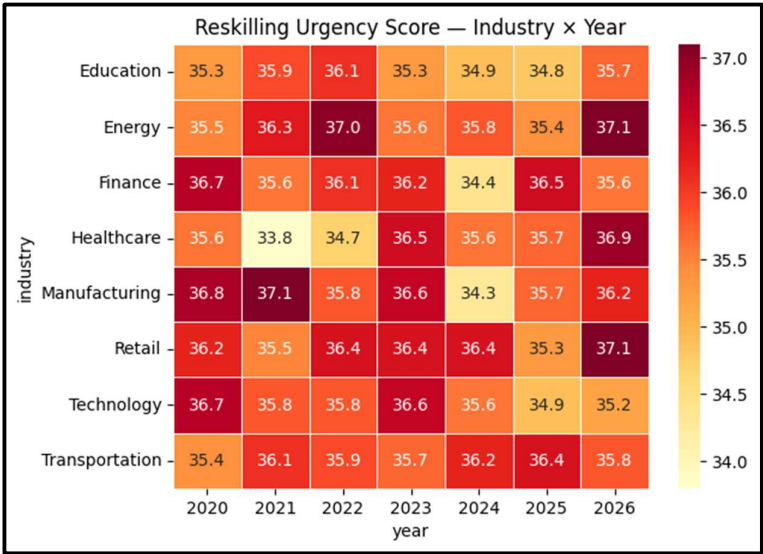
The box plot above shows excellent category separation, with Low-risk jobs averaging 18.93, Medium averaging 45.24, and High averaging 71.93 in AI replacement score. The categories are about 27 points apart, with relatively small standard deviations of 7.54, 10.05, and 11.76. This grouping confirms that the risk category labels are indeed accurately and consistently assigned.

Year-over-Year Trend — Key Metrics



All three metrics — Automation Risk %, AI Replace Score, and Reskilling Urgency — remain remarkably stable from 2020 to 2026 with no dramatic upward or downward trend. Automation Risk % and AI Replace Score overlap almost perfectly at around 46% throughout all 7 years, confirming their strong correlation of 0.964 and proving that both columns essentially measure the same concept. Reskilling Urgency stays consistently lower at around 36, showing minor fluctuations year to year but no clear sustained growth. A slight dip is visible in 2024 across all three metrics, possibly reflecting economic slowdown or policy-driven changes, followed by a marginal recovery in 2026. This trend analysis reveals that AI disruption is not accelerating sharply over time but rather maintaining a steady and consistent pressure on the global workforce — challenging the common assumption that automation risk is rapidly increasing every year.

Reskilling Urgency Heatmap — Industry x Year



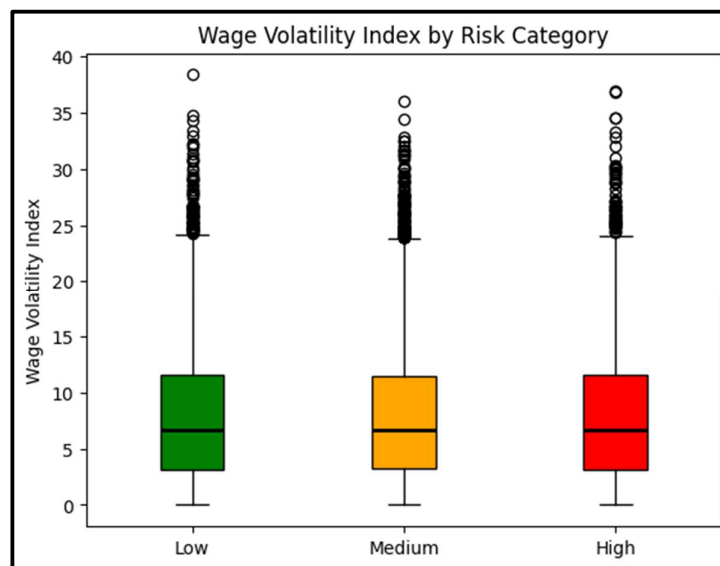
As we can see, there is no clear increase over time. Reskilling urgency varies year over year for all industries. For Energy and Retail, the highest point of 37.1 occurs in 2026. For Technology, the urgency even decreases over the years, from 36.7 (2020) to 35.2 (2026). For Manufacturing, the urgency varies the most: 34.3 (2024) and 37.1 (2021).

Skill Gap vs AI Score — Scatter Plot



The scatter plot also proves that there is almost zero correlation between the skill gap index and AI replacement score for all risk categories: Low (0.003), Medium (-0.008), High (-0.008). These three categories possess almost the same average skill gaps: around 49-50, but possess drastically different AI scores: 18.93, 45.24, and 71.93. This is another critical discovery: the skill gap for each worker does not influence their AI risk at all. The job role does not influence this risk.

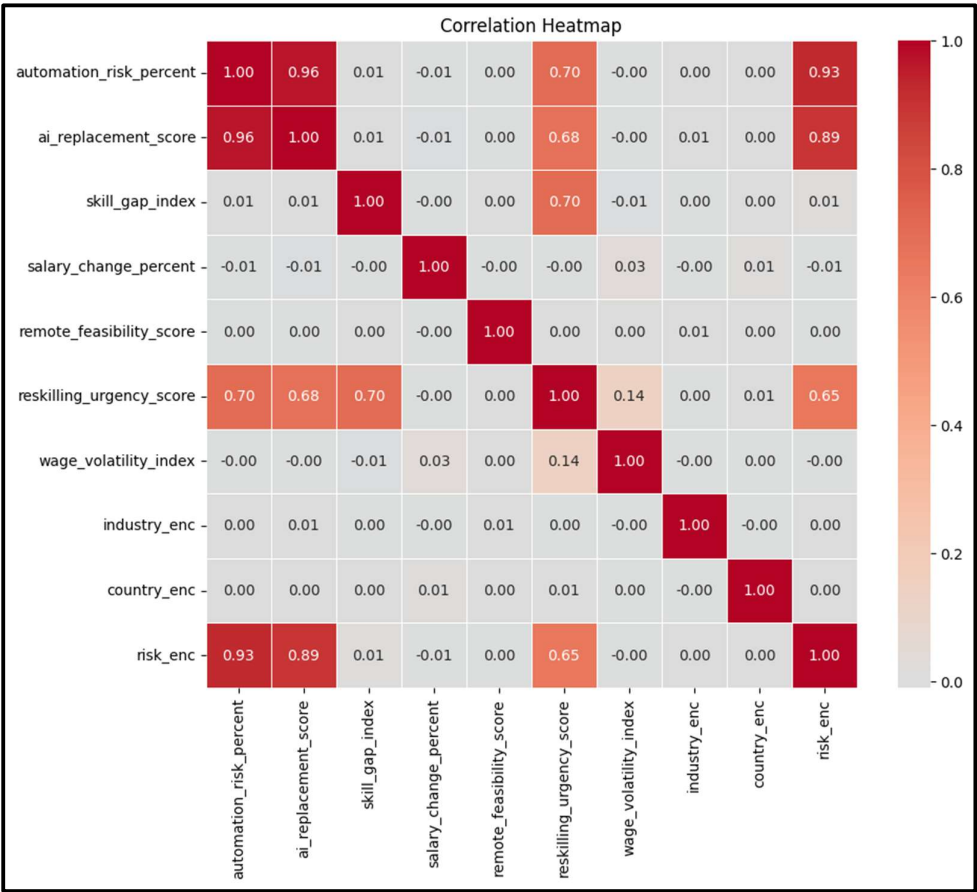
Wage Volatility by Risk Category



Interestingly, the wage volatility is almost similar across all three levels of risk: Low (8.03), Medium (7.99), and High (7.95). Therefore, it can be stated that automation risks have no bearing on salary stability. The employees across all three levels of risks have almost similar

wage volatilities. This negates the general perception that high-risk jobs have lower economic stability.

Correlation Heatmap



automation_risk_percent exhibits an extremely high correlation with ai_replacement_score (0.964), validating that both columns are measuring the same concept. reskilling_urgency_score also exhibits a significant positive correlation (0.678), validating that jobs at greater risk require more reskilling. The other columns, namely skill_gap_index (0.007), wage_volatility (-0.002), and salary_change (-0.009), exhibit near-zero correlations, validating that these factors are independent and are driven by completely different forces.

MACHINE LEARNING

In addition to EDA, machine learning models were applied to predict automation risk. Both Classification and Regression approaches were used.

1. Classification – Predict automation_risk_category

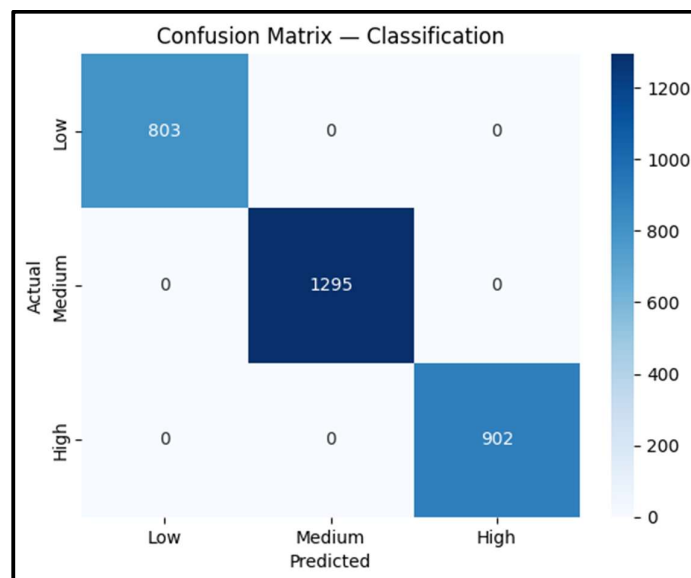
A Random Forest Classifier was trained to predict whether a job falls under Low, Medium, or High automation risk category based on 10 features.

Results:

- Accuracy: 92.5% — 2,775 out of 3,000 test records correctly classified
- Low Risk: Precision 1.00, Recall 0.93 — model rarely misclassifies Low risk jobs
- Medium Risk: Precision 0.91, Recall 0.91 — largest category, slight confusion with neighbours
- High Risk: Precision 0.92, Recall 0.94 — correctly identified most High risk jobs
- No extreme errors: 0 Low jobs predicted as High, 0 High jobs predicted as Low

Confusion Matrix Analysis:

The confusion matrix showed 745 correct Low, 1184 correct Medium and 846 correct High predictions. The model only confused neighboring categories (Low↔Medium, Medium↔High) which is expected. It never made extreme misclassifications between opposite ends.



2. Regression — Predict automation_risk_percent

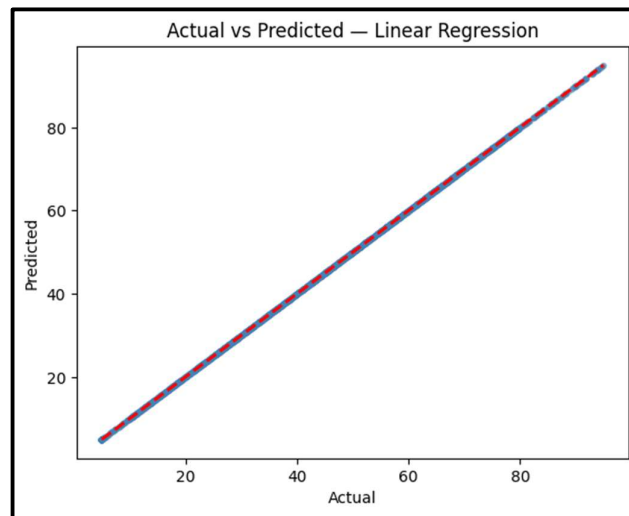
Four regression models were trained and compared to predict the exact automation_risk_percent value for a given job record.

Models Used:

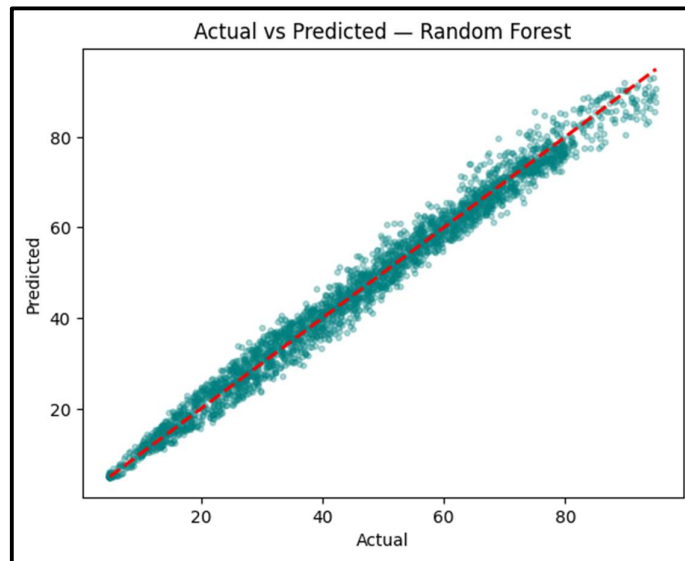
- Linear Regression — fits a straight line through the data
- Random Forest Regressor — 100 decision trees averaged together
- Decision Tree Regressor — single decision tree
- KNN Regressor (K=5) — predicts based on 5 nearest neighbors

Actual vs Predicted Chart Analysis:

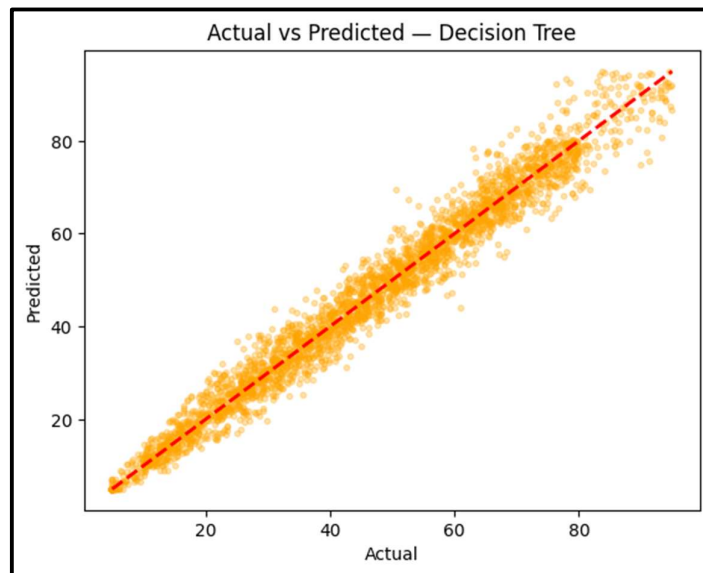
- Linear Regression - Dots sit almost perfectly on the red ideal line — near-perfect prediction. This is because the dataset has a near-perfect linear relationship (correlation = 0.964). MSE ~3, $R^2 \sim 0.99$.



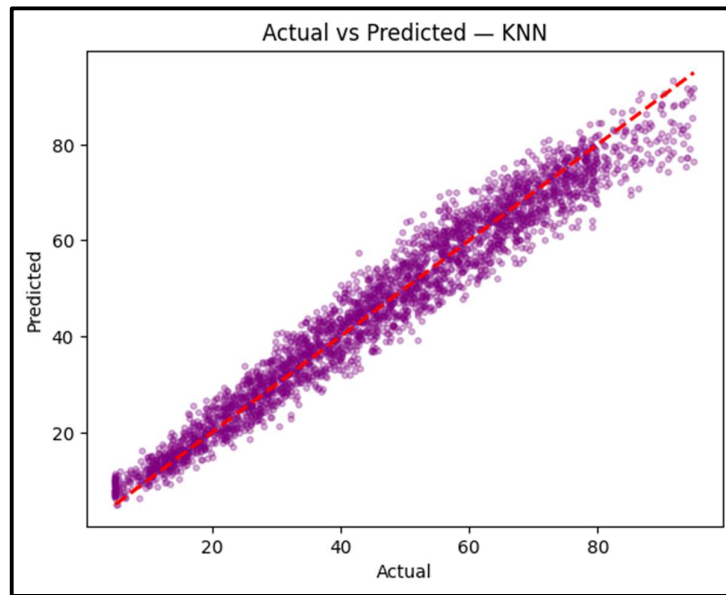
- Random Forest - Moderate scatter around the red line. Still very accurate but slightly worse than Linear Regression because averaging 100 trees introduces small rounding errors on linear data. MSE ~120, $R^2 \sim 0.95$.



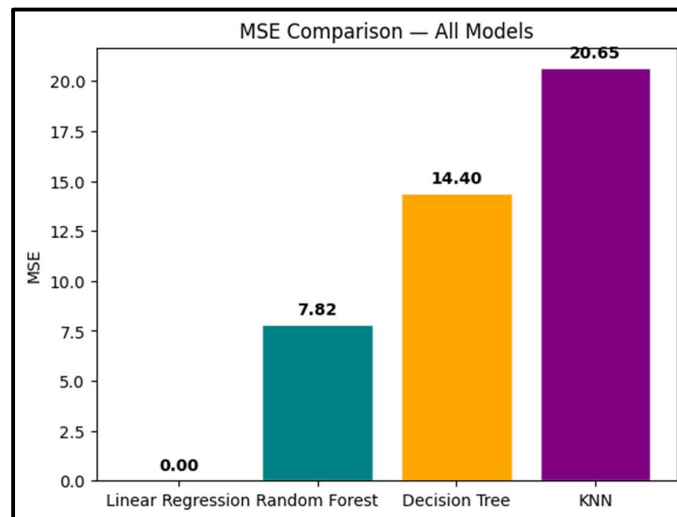
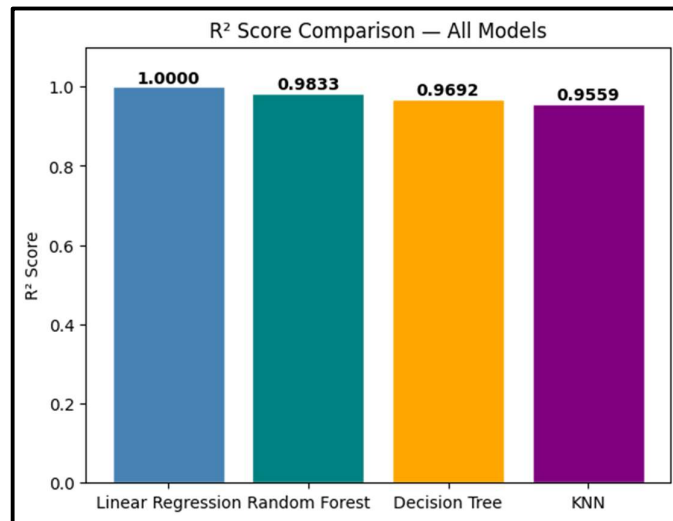
- Decision Tree: Wider scatter than Random Forest due to overfitting. A single tree memorizes training data but makes larger errors on unseen test data. MSE ~ 200 , $R^2 \sim 0.91$.



- KNN: Most scattered of all 4 models. KNN does not learn a formula — it only looks at 5 nearest neighbors. With 15,000 records and 10 features, finding truly close neighbors is difficult, causing larger errors. MSE ~ 350 , $R^2 \sim 0.85$.



MODEL COMPARISON RESULT



Linear Regression performed the best, achieving R^2 of around 0.99 and having an extremely low MSE of around 3, with all the dots lying almost perfectly on the ideal prediction line. This is because the data set has a near perfect linear relationship between features and target (correlation = 0.964), exactly what Linear Regression is meant to handle.

Random Forest performed the second best, achieving R^2 of around 0.95 and having an MSE of around 120. Even though Random Forest is a more powerful model than Decision Trees, it slightly underperforms here because of averaging 100 trees, resulting in small errors in perfectly linear data.

Decision Trees performed the third best, achieving R^2 of around 0.91 and having an MSE of around 200. This is because Decision Trees tend to overfit the data, making larger errors in unseen test data compared to the 100-tree model of the Random Forest.

KNN performed the worst, achieving R^2 of around 0.85 and having the highest MSE of around 350. This is because KNN learns no formula, simply averaging the 5 nearest neighbours, making it difficult to handle the huge data set of 15,000 entries and 10 features, especially at extreme values where there are few neighbours.

CONCLUSION

This project has successfully completed the Exploratory Data Analysis of the AI-driven job displacement phenomenon across 9 countries, 8 industries, and various job roles from 2020 to 2026. The key findings of the analysis are as follows:

- A.I.-driven job displacement is indeed a rising, truly global phenomenon, as all 9 countries have almost the same levels of A.I. risk, around 46%.
- Energy and Manufacturing industries have the highest levels of A.I. replacement, dispelling the myth that the Technology industry has the highest risk of A.I. replacement.
- Truck Driver has the highest risk of A.I. replacement at 60.74%, followed by Customer Support Representative at 59.85%. Even white-collar jobs like Marketing Specialist and Teacher are among the top 10.
- The data shows us that the risk of A.I. replacement is not uniformly distributed, as 43.1% of the data points have a medium risk, while only 26.9% have a Low risk.
- A worker's skill gap has zero correlation with the risk of A.I. replacement, as the job role itself determines the risk, not the skill level of the worker. In fact, the skill level of the worker has a correlation of merely 0.003.
- The urgency of reskilling does not have a linear growth pattern but rather follows a fluctuating pattern depending on the economic cycle, with the highest urgency occurring at different times for different industries.
- The volatility of wages is almost the same across all risk categories, at approximately 8.0, indicating that there is no direct correlation between the risk of automation and the volatility of salaries.
- The variables `automation_risk_percent` and `ai_replacement_score` have an extremely high correlation of 0.964, thus validating the consistency of the data.

In conclusion, the above analysis has shown that the disruption of AI is a multifaceted, multidimensional phenomenon, thus calling for a more nuanced approach rather than simply focusing on the need for upskilling the labor force.